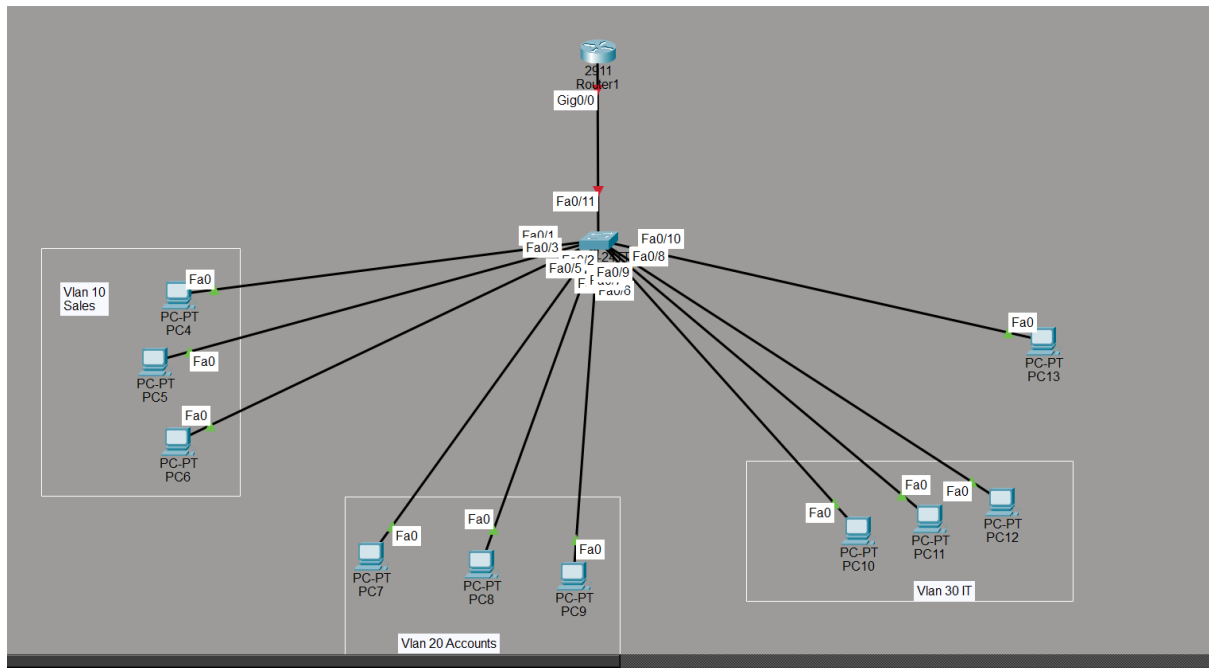


Switch and Router Configuration Basics

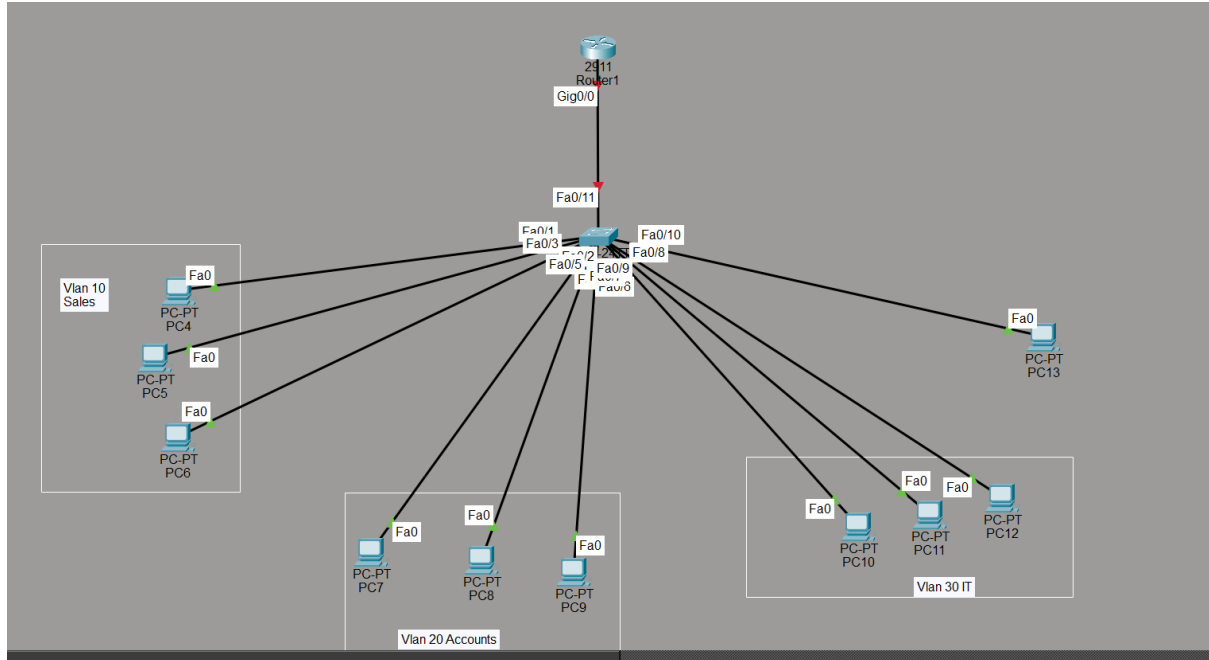
Task – 1 : Log in to Cisco Packet Tracer or any network simulation tool, add a switch, and create two VLANs named 'Sales' and 'Support' with VLAN IDs 10 and 20.

Solution :



Task – 2: Assign three different switch ports to VLAN 10 (Sales) and two ports to VLAN 20 (Support), then verify the VLAN membership using the show vlan command. Hint : Use the interface range command to select multiple ports efficiently.

Solution : Assign Ports to the VLANs



Commands :

```
interface range fastEthernet 0/1-3
```

```
switchport mode access
```

```
switchport access vlan 10
```

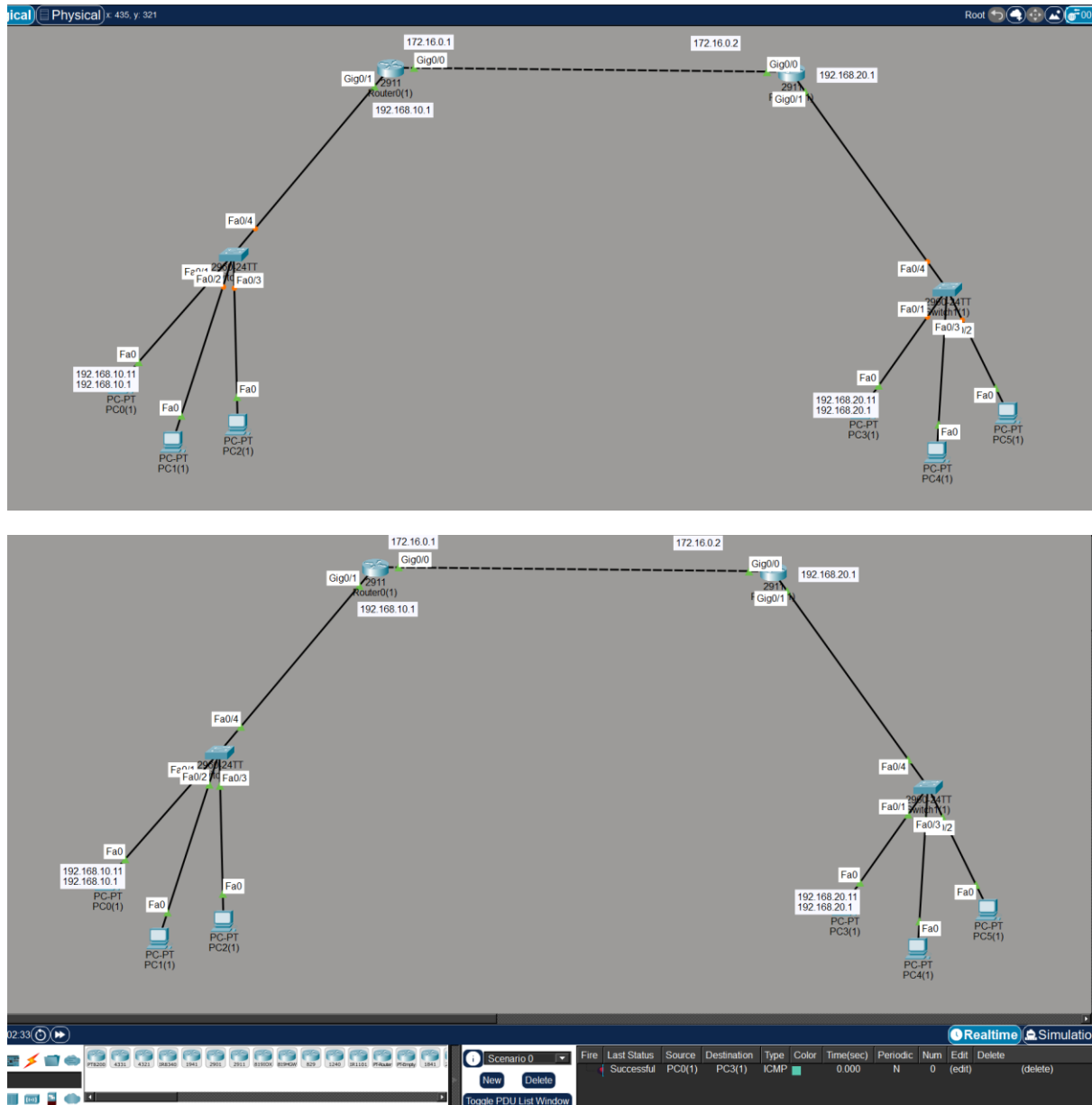
```
exit
```

```
end
```

```
copy running-config startup-config
```

Task – 3 : Connect two routers using a serial or Ethernet link in your simulation tool, assign IP addresses to their interfaces, and configure a static route on Router 1 to reach the LAN behind Router 2.

Solution : Configure a Static Route Between Two Routers



Task – 4 : Simulate a scenario where a device in the 'Sales' VLAN tries to ping a device in the 'Support' VLAN and explain why the ping fails, then suggest what configuration would allow inter-VLAN communication. Hint : Think about router-on-a-stick or Layer 3 switch concepts.

Solution : Sales VLAN → Support VLAN

Inter-VLAN Communication

Sales PC and Support PC are in different VLANs:

- **Sales PC:** 192.168.10.10, VLAN 10
- **Support PC:** 192.168.20.10, VLAN 20

A **Layer 2 switch cannot communicate between VLAN 10 and VLAN 20** because each VLAN is a separate broadcast domain.

To allow communication, we need a **Layer 3 device**, such as a router or Layer 3 switch. The router connects to the switch using a **trunk link** and provides a gateway for each VLAN.

- **VLAN 10:** Gateway 192.168.10.1
- **VLAN 20:** Gateway 192.168.20.1